

From stationary to event-driven, from static to dynamical, from classical to quantum — how should lens design adapt to the demands of Physical AI? And why? (April 27, 2026)

In an earlier post we asked: what is the lens design behind a physical-AI system that perceives the world by change, not by frames?

✦ The formal write-up is now drafted as a perspective:

◆ From Strehl to Kullback–Leibler — A Unified Fidelity Framework for Stationary, Event-Driven, and Quantum Imaging Optics.

■ The architecture this note highlights is three coupled axes:

✦ from classical to quantum

✦ from stationary to event-driven

✦ from static to dynamical

✦ The 2×2 typology — four reductions of one master functional on the eikonal bridge:

■ (C-S) Strehl

■ (C-E) Event-MTF

■ (Q-S) quantum state fidelity

■ (Q-E) modified fidelity (KL / Wasserstein-1).

✦ The numerical evidence:

■ Strehl decays roughly 2× faster than the normalized Event-MTF over the same wavefront-error budget.

■ Empirical scaling $F_e \propto S^\beta$ with $\beta \in [0.5, 0.7]$.

■ The choice of merit function is a choice of prescription.

✦ What's actionable today:

■ the practical reset ($PST < 1e-5$,

■ sub-0.1% coatings,

■ tighter CRA,

■ photon-counting QC) stays as posted.

■ New in the perspective: operand stubs for CODE V, Zemax, LightTools, and RSoft.

■ New merit-function operands, not new physics.

◆ Worth repeating, from the earlier post:

■ "single-photon detectors are already event-driven by their physics, and quantum photonics has been event-driven sensing and imaging from the start."

■ "We are only now learning to design the lenses for it."

👤 The perspective sits as supplementary material to the earlier post.

#PhysicalAI **#EventCamera** **#QuantumImaging** **#LensDesign** **#OpticalDesign** **#Photonics**